

SLC35G1: A highly chloride-sensitive transporter responsible for the basolateral membrane transport in intestinal citrate absorption

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eLife Assessment

This work identifies the molecular function of an orphan human transporter, SLC35G1, providing **convincing** evidence that this protein is involved in intestinal citrate absorption. This work provides **important** insight into transporter function and human physiology.

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Abstract

The intestinal absorption of essential nutrients, especially those not readily biosynthesized, is a critical physiological process for maintaining homeostasis. Numerous studies have indicated that intestinal absorption is mediated by various membrane transporters. Citrate, a crucial bioactive compound produced as an intermediate in the Krebs cycle, is absorbed in the small intestine through carrier-mediated systems because of its high hydrophilicity. While the luminal absorption of citrate is mediated by Na⁺-dicarboxylate cotransporter 1 (NaDC1/SLC13A2), the mechanism governing the release of the transported citrate into the bloodstream remains unknown. Here, we explored the transporters responsible for intestinal citrate absorption at the basolateral membrane, focusing on highly expressed orphan transporters in the small intestine as candidates. Consequently, SLC35G1, originally identified as a partner of stromal interaction molecule 1, a cell surface transmembrane glycoprotein, was found to play a role in the intestinal absorption of citrate at the basolateral membrane. Furthermore, our results revealed that SLC35G1-mediated citrate transport was diminished by chloride ions at physiologically relevant extracellular concentrations. This suggests that SLC35G1, to our best knowledge, is the first transporter identified to be extremely sensitive to chloride ions among those functioning on the basolateral membrane of intestinal epithelial

cells. This study provides valuable insights into the intestinal absorption of citrate and significantly contributes to elucidating the poorly understood molecular basis of the intestinal absorption system.

Introduction

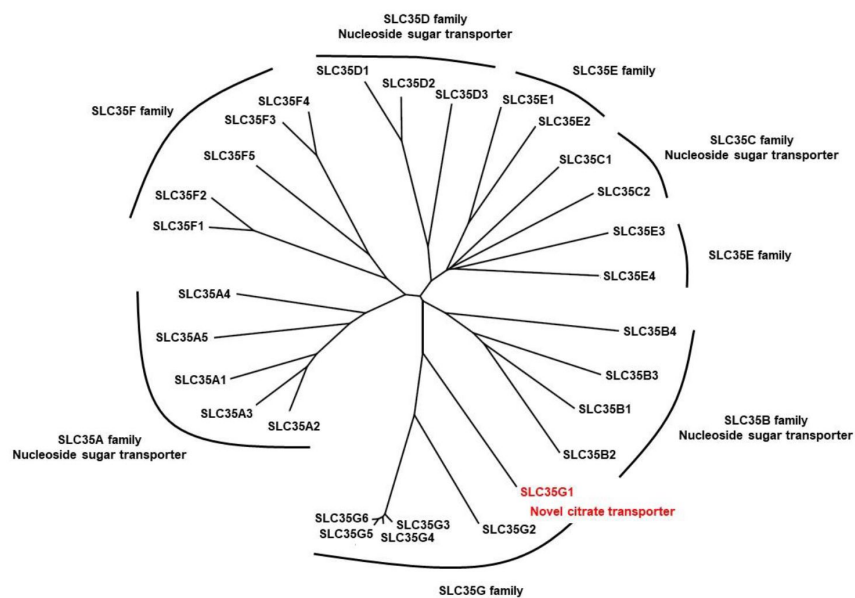
Citrate plays a crucial role as a tricarboxylic acid (TCA) cycle intermediate, contributing to the generation of metabolizable energy. Since citrate is obtained not only through biosynthesis but also from the diet, the existence of intestinal absorption system has long been acknowledged (Browne et al., 1978 [\[1\]](#)). Luminal absorption is predominantly mediated by Na⁺-dependent dicarboxylate transporter 1 (NaDC1/SLC13A2) (Pajor, 2006 [\[2\]](#)); however, the transporter responsible for the release of citrate and its metabolites across the basolateral membrane remain unidentified (Pajor, 2014 [\[3\]](#)). SLC62A1, which was initially recognized as a pyrophosphate transporter linked to joint mineralization, has recently been reported to facilitate citrate efflux in some cell types (Szeri et al., 2020 [\[4\]](#)). However, it is important to note that the role of SLC62A1 has been mainly implicated in osteoblasts, prostate, skeletal muscle, brain, and the cardiovascular system. It may have the major role in the systemic disposition of citrate rather than the intestinal absorption. To elucidate the molecular mechanism underlying citrate absorption at the basolateral membrane, we explored the transporters implicated in the process. Our study provides novel insights into basolateral membrane transporters, addressing a previously poorly understood aspect in intestinal epithelial cells.

Results and discussion

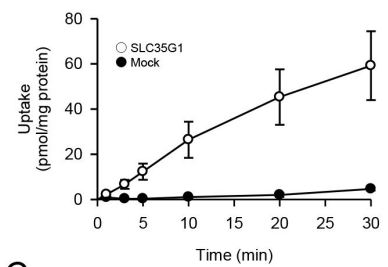
To identify a novel citrate transporter expressed in the small intestine, we conducted a comprehensive *in silico* screening to narrow down candidate orphan transporters. This initial screening involved bioinformatics analysis of gene expression database to identify candidate transporters that showed significant expression in intestine but were not or little characterized previously. Following the *in silico* screening, we cloned the candidate transporters, had them transiently expressed in human embryonic kidney 293 (HEK293) cells, and evaluated them for citrate transport activity by [¹⁴C]citrate uptake assay. Through this process, we discovered that SLC35G1, a member of the solute carrier (SLC) 35 family and originally identified as a partner of stromal interaction molecule 1 (Krapivinsky et al., 2011 [\[5\]](#)), is capable of transporting citrate at the plasma membrane. While the functions of nucleoside-sugar transporters on the endoplasmic reticulum membrane have been identified for certain SLC35 family members, including SLC35A, SLC35B, SLC35C, and SLC35D (Ishida and Kawakita, 2004 [\[6\]](#)), the roles of SLC35E, SLC35F, and SLC35G have not yet been fully elucidated, as shown in **Figure 1A** [\[7\]](#). SLC35G1 was indicated to be mainly localized to the endoplasmic reticulum (ER) in the earlier study, in which SLC35G1 was tagged with GFP. A possibility is that SLC35G1 was wrongly directed to ER due to the modulation in the study. For the functional analysis, we first established Madin-Darby canine kidney (MDCKII) cells stably expressing SLC35G1. Using these cells, we observed a linear increase in SLC35G1-mediated citrate uptake with time, specifically during the first 10 min in Hanks' solution mimicking the cytosolic ion content by eliminating Cl⁻ (**Figure 1B** [\[8\]](#)). The acidic condition (pH 5.5) was used to gain sufficient uptake for functional analysis. Therefore, the uptake at 10 min was evaluated in subsequent studies. A kinetic analysis revealed that the SLC35G1-specific uptake of citrate exhibited saturation kinetics, with a $V_{\max}$ of 1.10 nmol/min/mg protein and a K_m of 519 μ M (**Figure 1C** [\[9\]](#)). Notably, the specific uptake of citrate by SLC35G1 was extensively inhibited by extracellular Cl⁻ (**Figure 1D** [\[10\]](#)), with an IC_{50} value of 6.7 mM (**Figure 1E** [\[11\]](#)). As previously reported (Melkikh and Sutormina, 2008 [\[12\]](#)), the Cl⁻ concentration is lower in the cytosol (5 mM) than in the extracellular conditions (120 mM). Therefore, SLC35G1 could be a potential citrate exporter. It is notable that, as shown in **Figure 1D** [\[13\]](#), the use of K-gluconate in place of Na-

gluconate, which induces plasma membrane depolarization, had no impact on the specific uptake, suggesting that SLC35G1-mediated citrate transport is independent of membrane potential. Additionally, we examined the effect of various compounds (200 μ M), to explore the potential role of SLC35G1 in their transport (**Figure 1F**). The tested compounds were TCA cycle intermediates (fumarate, α -ketoglutarate, malate, pyruvate, and succinate) as substrate candidate carboxylates analogous to citrate, diverse anionic compounds (BSP, DIDS, probenecid, pravastatin, and taurocholate) as those that might be substrates or inhibitors, and diverse cationic compounds (cimetidine, quinidine, and verapamil) as those that are least likely to interact with SLC35G1. Among them, certain anionic compounds significantly reduced SLC35G1-specific citrate uptake, suggesting that they may interact with SLC35G1. However, we could not identify any structural features commonly shared by these compounds, except that they have anionic moieties. At the concentration of 200 μ M, competing substrates with an affinity comparable to that of citrate could inhibit SLC35G1-specific citrate uptake by about 30%. Therefore, it is likely that the compounds that did not exhibit significant effect have no affinity or at least lower affinity than citrate to SLC35G1. SLC35G1-specific citrate uptake showed pH dependence (**Figure 1G**). To investigate whether SLC35G1 utilizes an H^+ gradient as a driving force for citrate transport, we examined the effect of the protonophore nigericin in the uptake solution at pH 5.5 to dissipate the H^+ gradient across the plasma membrane. However, treatment with nigericin did not affect the specific uptake of citrate (**Figure 1H**), excluding this possibility. Based on these, it is likely that SLC35G1 is a facilitative transporter, although we cannot exclude another possibility that its sensitivity to extracellular Cl^- might imply its operation as a citrate/ Cl^- exchanger. Given that one carboxylate can be protonated with a pK_a of 5.4 (Contreras-Trigo et al., 2018) (**Figure 1I**), SLC35G1 may preferably transport dianionic citrate, like NaDC1 (Pajor and Sun, 1996). Finally, we determined the role of SLC35G1 in the transcellular transport of citrate at the basolateral membranes using SLC35G1-transfected polarized MDCKII cells on Transwell permeable membrane filters. As shown in **Figure 1J**, SLC35G1 predominantly localized to the basolateral membrane of polarized MDCKII cells. Transport studies were performed under pH gradient conditions (apical pH 5.5 and basal pH 7.4) in the presence of Cl^- (144 mM) to mimic intestinal physiological conditions. SLC35G1-transfected cells showed no significant change in intracellular citrate accumulation from the apical side, compared to mock cells (**Figure 1K**). However, apical-to-basolateral transcellular transport was significantly higher in SLC35G1-transfected cells (**Figure 1L**). On the other hand, SLC35G1 overexpression affected neither intracellular accumulation nor transcellular transport from the basolateral side. This is reasonable since the basolaterally localized SLC35G1 operates for efflux but not for uptake in the presence of Cl^- in the extracellular environment. These results suggest that SLC35G1 plays a role in the intestinal absorption of citrate at the basolateral membrane as an exporter under physiological conditions.

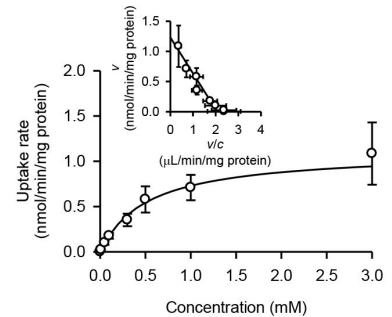
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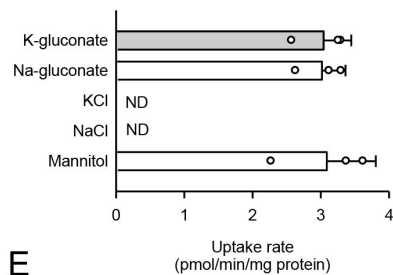
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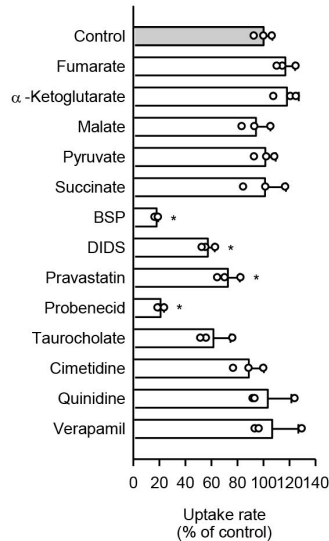
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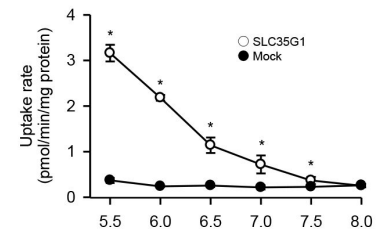
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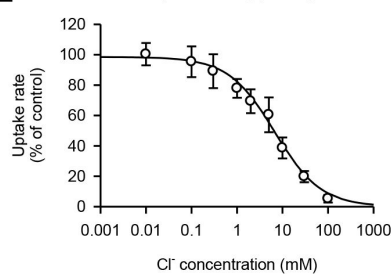
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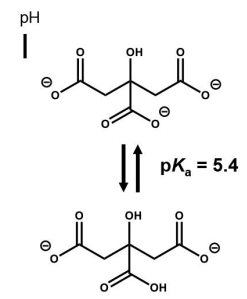
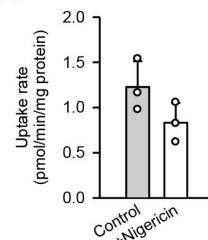
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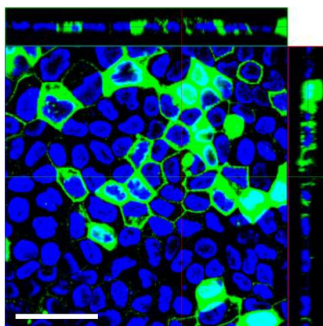
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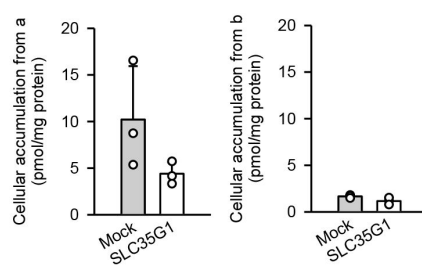
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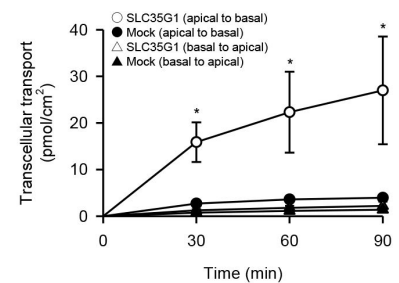


Figure 1.

Functional characteristics of SLC35G1 stably expressed in MDCKII cells. (A) Phylogenetic tree of the SLC35 members and identified functions. (B) Time course of [^{14}C]citrate (1 μM) uptake in SLC35G1-transfected MDCKII cells (open circles) and mock cells (closed circles), evaluated at pH 5.5 and 37°C under the Cl^- -free condition. (C) Concentration-dependent [^{14}C]citrate uptake by SLC35G1, evaluated at various concentrations for 10 min at pH 5.5 and 37°C under the Cl^- -free condition. The estimated values of V_{max} and K_m were 1.11 ± 0.16 nmol/min/mg protein and 519 ± 86 μM , respectively. (D) Effect of ionic conditions on [^{14}C]citrate (1 μM) uptake by SLC35G1, evaluated for 10 min at pH 5.5 and 37°C. K-gluconate in the control uptake solution was replaced as indicated. (E) Concentration-dependent chloride inhibition of [^{14}C]citrate (1 μM) uptake by SLC35G1, evaluated at various chloride concentrations for 10 min at pH 5.5 and 37°C. The estimated value of IC_{50} was 6.7 ± 1.4 mM. (F) Effect of various compounds on [^{14}C]citrate (1 μM) uptake by SLC35G1, evaluated for 10 min at pH 5.5 and 37°C in the presence (200 μM) or absence (control) of a test compound under the Cl^- -free condition. BSP, bromosulphophthalein; DIDS, 4, 4'-diisothiocyano-2, 2'-stilbenedisulfonic acid. (G) Effect of pH on [^{14}C]citrate (1 μM) uptake in SLC35G1-transfected MDCKII cells (open circles) and mock cells (closed circles), evaluated for 10 min at 37°C under the Cl^- -free condition, using the uptake solutions supplemented with 10 mM MES (pH 6.5 and below) or 10 mM HEPES (pH 7.0 and above). (H) Effect of the protonophore nigericin on [^{14}C]citrate (1 μM) uptake by SLC35G1, evaluated for 10 min at pH 5.5 and 37°C under the Cl^- -free condition in the presence (10 μM) or absence (control) of nigericin after pretreatment for 5 min with or without nigericin under the same conditions. (I) Structural formula illustrating the chemical equilibrium of citrate. (J) Cellular localization of SLC35G1 stably expressed in polarized MDCKII cells. The confocal laser-scanning microscopic image shows SLC35G1 (green) and nuclei stained with 4', 6'-diamidino-2-phenylindole (DAPI; blue). Scale bar, 50 μm . (K) Intracellular accumulation of citrate in polarized MDCKII cells stably expressing SLC35G1 and mock cells on Transwell filters. Intracellular accumulation of [^{14}C]citrate (1 μM) was evaluated for 90 min at pH 5.5 on the apical side (a) and pH 7.4 on the basolateral side (b) in the presence of Cl^- (144 mM) on both sides of intestinal epithelial cells. (L) Time course of transcellular [^{14}C]citrate (1 μM) transport in polarized MDCKII cells stably expressing SLC35G1 and mock cells on Transwell filters, evaluated at pH 5.5 in the apical chamber and pH 7.4 in the basolateral chamber in the presence of Cl^- (144 mM) in both chambers. Data represent the mean $\pm$ SD of three biological replicates (B-H, K, L). Statistical differences were assessed using analysis of variance (ANOVA) followed by Dunnett's test (D, F, L) or using Student's t-test (G, H, K). *, $p < 0.05$ compared with the control (F), or compared with the mock at each pH (G) or time point (L).

To further investigate the physiological role of SLC35G1, we examined its expression in human tissues (Figure 2A). Quantitative real-time PCR analysis revealed that SLC35G1 is highly expressed in the digestive tract, especially in the upper part of the small intestine, including the duodenum and jejunum, followed by the testis and pancreas. Additionally, immunohistochemical staining of the human jejunum clearly showed that SLC35G1 is localized on the basolateral membrane, as indicated by the co-localization of ATP1A1, a marker protein on the basolateral membrane (Figure 2B). To investigate the physiological impact of SLC35G1 on intestinal citrate absorption, we examined the effect of SLC35G1 knockdown on citrate uptake in human colorectal adenocarcinoma Caco-2 cells, a commonly used model for studying intestinal absorbency (Yee, 1997) (Figure 2C), in the absence of Cl^- . As shown in Figure 2C, citrate uptake was significantly reduced following RNA interference with siRNAs targeting SLC35G1 mRNA, providing evidence of the involvement of SLC35G1 in intestinal citrate absorption. The reduction in citrate uptake was not indicative of off-target effects, as multiple siRNAs with distinct nucleic acid sequences yielded similar reductions. Moreover, the expression of SLC35G1 mRNA was reduced by these siRNAs, confirming successful silencing of SLC35G1 (Figure 2D). The correlation between the decrease in citrate uptake and mRNA levels suggests that SLC35G1 is primarily involved in citrate uptake. Immunohistochemical staining revealed that endogenous SLC35G1 in polarized Caco-2 cells cultured on Transwell membranes was also localized on the basolateral membranes (Figure 2E). Taken together, these results indicate that intestinal citrate transport at the

basolateral membrane is primarily mediated by SLC35G1. Additionally, it acts cooperatively with NaDC1, which is reported to be highly expressed in the jejunum (Pajor, 1995 [↗](#)), for citrate absorption on the apical membrane, as illustrated in **Figure 2F** [↗](#).

Our findings suggest that SLC35G1 is localized on the basolateral side of intestinal epithelial cells and, to our knowledge, is the first transporter identified with chloride-sensitivity, which is a unique characteristic to render with the directional transport function for an SLC transporter. These findings pave the way for several follow-up directions related to the molecular basis of the intestinal absorption of various compounds, including nutrients and drugs.

Materials and Methods

Materials

[¹⁴C]citrate (116.4 mCi/mmol) was obtained from PerkinElmer Life and Analytical Sciences (Boston, MA, USA). Unlabeled citrate and Dulbecco's modified Eagle's medium (DMEM) were sourced from Wako Pure Chemical Industries (Osaka, Japan), and fetal bovine serum (FBS) was obtained from Sigma Aldrich (St. Louis, MO, USA). All other reagents were of analytical grade and commercially acquired.

Cell culture

Madin-Darby canine kidney II (MDCKII) cells and Caco-2 cells were cultured at 37°C and 5% CO₂ in DMEM supplemented with 10% FBS, 100 U/mL penicillin, and 100 µg/mL streptomycin, as previously described (Mimura et al., 2017 [↗](#)).

Preparation of plasmids

The cDNA for human SLC35G1 was cloned using a reverse transcription (RT)-PCR-based method, as previously described (Mimura et al., 2017 [↗](#)). Briefly, an RT reaction was performed to obtain a cDNA mixture, using 1 µg of the human intestine total RNA (BioChain Institute, Newark, CA), an oligo(dT) primer, and the highly efficient reverse transcriptase ReverTra Ace (Toyobo, Osaka, Japan). The SLC35G1 cDNA was then amplified by PCR, using KOD plus polymerase (Toyobo) and the following primers: forward primer, 5'-GAGATGCGGCCTCAGGACAG-3', and reverse primer, 5'-TAGCCTCCCCACCCATATCCC-3'. These primers were designed based on the GenBank sequences for SCL35G1 (accession number: NM_001134658.2). The second PCR was performed using the initial PCR product as a template, with a forward primer containing an EcoRI restriction site (underlined), 5'-AGAATTCCGAGATGCGGCCTCAGGAC-3', and a reverse primer containing an XbaI restriction site (underlined), 5'-GTTCTAGATGTGTGCGTATGCTG-3'. The resulting cDNA product was inserted into the pCI-neo vector (Promega, Madison, WI, USA) between the EcoRI and XbaI sites, and the sequence of the final product was determined using an automated sequencer.

Uptake study

First, we established MDCKII cells stably expressing SLC35G1 as previously described (Yamamoto et al., 2010 [↗](#)). Briefly, MDCKII cells were transfected with the plasmid carrying the SLC35G1 cDNA, using Lipofectamine 2000 as a transfection reagent. The cells were then cultured in DMEM supplemented with 10% FBS and 800 µg/mL G418 for 2–3 weeks. Subsequently, G418-resistant clones were selected and assessed for the transport of [¹⁴C]citrate. MDCKII cells stably expressing SLC35G1 with high citrate transport activity were used in subsequent studies.

MDCKII cells stably expressing SLC35G1 (1.5×10^5 cells/mL, 1 mL/well) were grown in 24-well plates for 48 h until they reached confluence. For standard transport assays, cells were preincubated in substrate-free uptake buffer, which was Hanks' solution modified to reflect

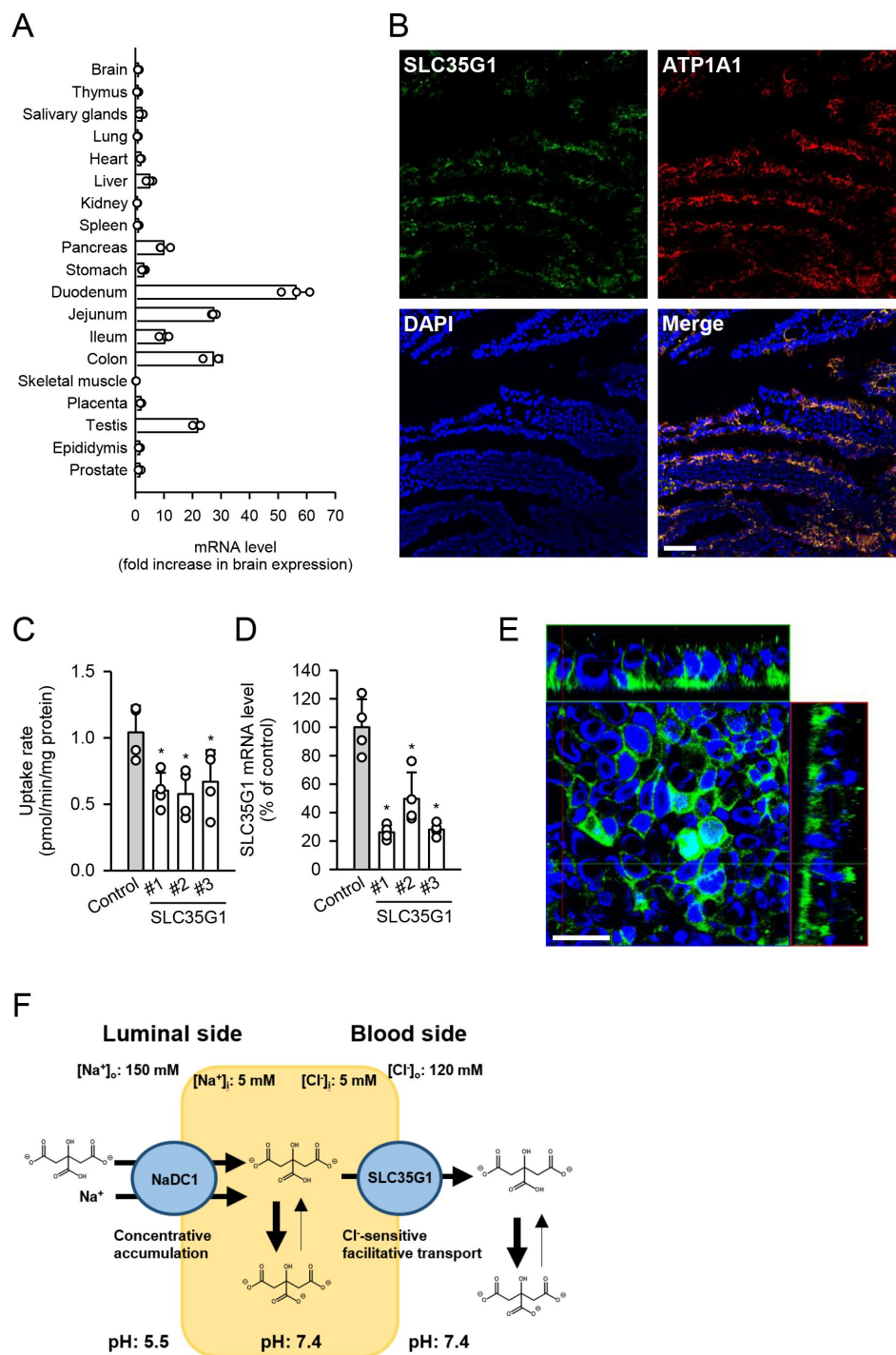


Figure 2.

(A) Analysis of SLC35G1 mRNA expression in various human tissues using quantitative real-time PCR. (B) The immunofluorescent images show the subcellular localization of SLC35G1 (green), ATP1A1 (red), and 4', 6-diamidino-2-phenylindole (DAPI; blue) in the human jejunum. Scale bar, 100 μ m. (C-D) Effect of SLC35G1 silencing on citrate uptake and on SLC35G1 mRNA expression in Caco-2 cells. (C) The uptake rate of [14 C]citrate (1 μ M) was evaluated for 10 min at pH 5.5 and 37°C under the Cl⁻-free condition in Caco-2 cells transfected with three different siRNAs specific to SLC35G1 mRNA or control siRNA. (D) SLC35G1 mRNA levels were assessed using quantitative real-time PCR analysis. (E) Cellular localization of endogenous SLC35G1 in polarized Caco-2 cells. The confocal laser-scanning microscopic image shows SLC35G1 (green) and nuclei stained with DAPI (blue). Scale bar, 50 μ m. (F) Schematic model showing intestinal citrate absorption mediated by NaDC1 and SLC35G1. Data represent the mean $\pm$ SD of three technical repeats (A) or biological repeats (C-D). Statistical differences were assessed using analysis of variance (ANOVA) followed by Dunnett's test (C-D).

cytosolic conditions (142.06 mmol/L K-gluconate, 0.812 mmol/L MgSO₄, 0.379 mM K₂HPO₄, 0.441 mM KH₂PO₄, 0.952 mM Ca-gluconate, 25 mM D-glucose) and supplemented with 10 mM 2-(N-morpholino)-ethanesulfonic acid (MES; pH 5.5), for 5 min. Subsequently, uptake assays were initiated by replacing the substrate-free uptake buffer with an uptake buffer containing a [¹⁴C]citrate (0.25 mL). For initial uptake analysis experiments, the uptake period was set to 10 min in the initial phase, in which uptake was in proportion to time. Other experiments were conducted in a time-dependent manner, as specified. When examining the effects of various compounds on citrate uptake, the test compounds were added to the buffer only during the uptake period. All procedures were performed at 37°C. Assays were stopped by adding 2 mL of ice-cold substrate-free uptake buffer, followed by two washes with an additional 2 mL of the same buffer. The cells were solubilized in 0.5 mL of 0.2 mol/L NaOH solution containing 0.5% sodium dodecyl sulfate (SDS), and the associated radioactivity was measured by liquid scintillation counting for uptake evaluation. The cellular protein content was determined by the bicinchoninic acid (BCA) method, using bovine serum albumin as the standard (3). In experiments to examine the effect of ionic conditions, K-gluconate in the control uptake solution was replaced as indicated. Uptake solutions were supplemented with 10 mM MES (pH 6.5 and below) or 10 mM HEPES (pH 7.0 and above) in experiments to examine the effect of pH. In experiments to examine the effect of nigericin as an agent for intracellular acidification, a mannitol-based buffer (250 mM mannitol, 1.2 mM MgSO₄, 2 mM KH₂PO₄, 5 mM D-glucose, 10 μM nigericin, and 20 mM MES) was used and uptake assays were conducted for 10 min at pH 5.5 and 37°C in the presence (10 μM) or absence (control) of nigericin after pretreatment for 5 min with or without nigericin under the same conditions. To estimate nonspecific uptake, uptake assays were conducted in mock cells transfected with an empty pCI-neo vector. The specific uptake of citrate by SLC35G1 was estimated by subtracting the uptake in mock cells from that in SLC35G1-transfected cells.

Transcellular transport study

MDCKII cells were seeded at a density of 2×10^5 cells on each polycarbonate membrane insert in a 12-well Transwell plate and cultured for 5 days. After removing the culture medium from both sides of the inserts, cells were preincubated for 5 min at 37°C in Hanks' solution (136.7 mM NaCl, 5.36 mM KCl, 0.952 mM CaCl₂, 0.812 mM MgSO₄, 0.441 mM KH₂PO₄, 0.385 mM Na₂HPO₄, and 25 mM D-glucose) supplemented with 10 mM MES (pH 5.5) in the apical chamber and 10 mM HEPES (pH 7.4) in the basolateral chamber. To initiate transcellular transport from the apical side to the basolateral side, 0.5 mL of the Hanks' solution (pH 5.5) containing [¹⁴C]citrate was replaced in the apical chamber. For transport in the opposite direction (basolateral side to apical side), 1.5 mL of Hanks' solution (pH 7.4) containing [¹⁴C]citrate was replaced in the basal chamber. To monitor transcellular transport, samples (100 μL from basal chamber and 50 μL from the apical chamber) were periodically collected, replenishing each collected volume with an equal volume of fresh buffer. At the end of the transport study, the assays were stopped by adding ice-cold Hanks' solution (1 mL for the apical chamber and 3 mL for the basolateral chamber), followed by washing the cells twice with the same buffer. The cells were then solubilized in 0.5 mL of 0.2 M NaOH solution containing 0.5% SDS at room temperature for 1 h. The radioactivity associated with the cells was measured by liquid scintillation counting for uptake evaluation.

Quantification of SLC35G1 mRNA by quantitative real-time PCR

Total RNA samples from various human tissues (BioChain Institute) and Caco-2 cells were used to prepare cDNA using the ReverTra Ace reverse transcriptase. A quantitative real-time PCR was performed using SsoFast EvaGreen Supermix with Low ROX (Bio-Rad Laboratories, Hercules, CA, USA) on a 7300 Fast Real-time PCR System (Applied Biosystems, Foster City, CA, USA) with the following primers: forward primer for SLC35G1, 5'-AGAGCCCACTGAGAAAAGGA-3'; reverse primer for SLC35G1, 5'-GTAGGTGCTGGCTCTGCCT-3'; forward primer for GAPDH, 5'-CGGAGTCAACGGATTGGTCGTAT-3'; reverse primer for GAPDH, 5'-AGCCTTCTCCATGGTGGTGAAGAC-3'. GAPDH was employed as an internal control for normalization of the mRNA expression levels of SLC35G1.

Knockdown study

Caco-2 cells (5.0×10^4 cells/mL, 1 mL/well) were grown on 24-well-coated plates for 6 h, transfected with 5 pmol/well of the Silencer Selected siRNAs specific to the mRNA of SLC35G1 (Thermo Fisher Scientific, Waltham, MA), using 1.5 mL/well of Lipofectamine RNAi MAX (Thermo Fisher Scientific), and cultured for 5 days for silencing of the designated transporter by RNAi. The sequences of the siRNAs are following: sense sequence of #1 siRNA, GGAGUGAUCCUUAUCGUGATT; antisense sequence of #1 siRNA, UCACGAUAAGGAUCACUCCAG, sense sequence of #2 siRNA, CUUGC UUAUAUACAGAAATT; antisense sequence of #2 siRNA, UUUCUGUAUAUUAAGCAAGGG, sense sequence of #3 siRNA, CAUUUGGUAUUAUGUAGUATT; antisense sequence of #3 siRNA, UACUACAUAUACCAAUGCT. For control, negative control Silencer Selected siRNA (Thermo Fisher Scientific), was used. Citrate uptake assays were conducted as described above for SLC35G1-transfected MDCKII cells.

Immunofluorescence staining

To examine the localization of SLC35G1, MDCKII cells stably expressing SLC35G1 were seeded at a density of 2×10^5 cells/insert on a 12-well Transwell plate and cultured for 5 days, and Caco-2 cells were seeded at a density of 5×10^5 cells/insert on a 12-well Transwell plate and cultured for 21 days. The cells were washed thrice with ice-cold PBS and fixed in 5% paraformaldehyde for 15 min. After washing thrice with PBS, the cells were incubated for 1 h at room temperature with an anti-human SLC35G1 goat polyclonal antibody at a dilution of 1:100 in Can Get Signal Immunostain Immunoreaction Enhancer Solution A (Toyobo). The cells were washed with PBS, and the primary antibody was probed with anti-goat IgG Alexa Fluor 488 (Jackson ImmunoResearch Laboratories, West Grove, PA, USA) at a dilution of 1:400 in Can Get Signal Immunostain Immunoreaction Enhancer Solution B (Toyobo) for 1 h at room temperature. After three washes with PBS, the cells were mounted on a glass slide in 9:1 glycerol/PBS containing 1 μ M DAPI.

A specimen of normal human adult small intestine frozen tissue sections (US Biomax, Derwood, MD, USA) was blocked in G-Block (GenoStaff, Tokyo, Japan) for 1 h and then incubated for 1 h at room temperature with an anti-human SLC35G1 goat polyclonal antibody at a dilution of 1:100 in Can Get Signal Immunostain Immunoreaction Enhancer Solution A. After washing thrice with PBS, the primary antibody was probed with anti-goat IgG Alexa Fluor 488 by incubating for 1 h at room temperature.

Additionally, we stained ATP1A1 with an anti-human ATP1A1 mouse monoclonal antibody (clone BXP-21, Abcam, Cambridge, UK) labeled with the Ab-10 Rapid Hilyte Fluor 647 labeling kit (Dojindo Laboratories, Kumamoto, Japan) according to the manufacturer's instructions. After washing thrice with PBS, the tissues were mounted in 9:1 glycerol/PBS containing 1 μ M DAPI.

The localization of immunofluorescently labeled proteins and nuclei was visualized using a confocal laser scanning microscope (LSM510; Zeiss, Jena, Germany).

Data Analysis

The saturable transport of citrate by SLC35G1 was analyzed by assuming Michaelis-Menten-type carrier-mediated transport, as represented by the following equation: $v = V_{\max} \times s / (K_m + s)$. The apparent parameters of the maximum transport rate ($V_{\max}$) and Michaelis constant (K_m) were estimated by fitting this equation to the experimental profile of the uptake rate (v) versus the concentration (s) of citrate as the substrate, using a nonlinear least-squares regression analysis program, WinNonlin (Pharsight, Mountain View, CA, USA), with v^{-2} as the weight.

When s is considerably smaller than K_m ($s \ll K_m$), v in the presence of an inhibitor can be expressed as $v = v_0 / (1 + (i/IC_{50})^n)$. The half-maximal inhibitory concentration (IC_{50}) was estimated, along with the Hill coefficient (n) and v in the absence of inhibitors (v_0), by fitting this equation to the experimental profile of v versus the inhibitor concentration (i).

Acknowledgements

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Editors

Reviewing Editor

Randy Stockbridge

University of Michigan, Ann Arbor, United States of America

Senior Editor

Merritt Maduke

Stanford University, Stanford, United States of America

Reviewer #1 (Public review):

Summary:

The current manuscript provides solid evidence that the molecular function of SLC35G1, an orphan human SLC transporter, is citrate export at the basolateral membrane of intestinal epithelial cells. Multiple lines of evidence, including radioactive transport experiments, immunohistochemical staining, gene expression analysis, and siRNA knockdown are combined to deduce a model of the physiological role of this transporter.

Strengths:

The experimental approaches are comprehensive, and together establish a strong model for the role of SLC35G1 in citrate uptake. The observation that chloride inhibits uptake suggests an interesting mechanism that exploits the difference in chloride concentration across the basolateral membrane.

Weaknesses:

A gap in this study is that the mechanism of the transporter has not been established. The authors propose that the mechanism is facilitated diffusion, while also leaving open the possibility that citrate transport is coupled to another ion, such as chloride. However, another result from this study seems to be in conflict with the proposed facilitative diffusion mechanism. Specifically, the study finds that uptake is not impacted by membrane depolarization. This would imply that transport is not electrogenic, whereas facilitated diffusion of citrate anion should be an electrogenic process.

<https://doi.org/10.7554/eLife.98853.2.sa3>

Reviewer #2 (Public review):

Summary:

The primary goal of this study was to identify the transport pathway that is responsible for the release of dietary citrate from enterocytes into blood across the basolateral membrane.

Strengths:

The transport pathway responsible for the entry of dietary citrate into enterocytes was already known, but the transporter responsible for the second step remained unidentified.

The studies presented in this manuscript identify SLC35G1 as the most likely transporter that mediates the release of absorbed citrate from intestinal cells into the serosal side. This fills an important gap in our current knowledge on the transcellular absorption of dietary citrate. The exclusive localization of the transporter in the basolateral membrane of human intestinal cells and the human intestinal cell line Caco-2 and the inhibition of the transporter function by chloride support this conclusion.

Weaknesses:

(i) The substrate specificity experiments have been done with relatively low concentrations of potential competing substrates, considering the relatively low affinity of the transporter for citrate. Given that NaDC1 brings in not only citrate as a divalent anion and also other divalent anions such as succinate, it is possible that SLC35G1 is responsible for the release of not only citrate but also other dicarboxylates. However the substrate specificity studies show that the dicarboxylates tested did not compete with citrate, meaning that SLC35G1 is selective for the citrate (2-), but this conclusion might be flawed because of the low concentration of the competing substrates used in the experiment. Furthermore, the apical NaDC1 is not selective for citrate; in fact, it transports citrate with a much lower affinity than it transports dicarboxylates such as succinate. If what the authors suggest that SLC35G1 is selective for citrate is correct, there must be another transporter for the efflux of dicarboxylates. The authors should have performed a dose-response experiment for the dicarboxylates tested as potential substrates before making the conclusion that SLC35G1 is selective for citrate.

(ii) The authors have used MDCK cells for assessment of the transcellular transfer of citrate via SLC35G1, but it is not clear whether this cell line expresses NaDC1 in the apical membrane as the enterocytes do. Even though the authors expressed SLC35G1 ectopically in MDCK cells and showed that the transporter localizes to the basolateral membrane, the question as to how citrate actually enters the apical membrane for SLC35G1 in the other membrane to work remains unanswered.

(iii) The role of chloride in the efflux of citrate remains not evaluated in detail. Similarly, the potential role of membrane potential in the transport function of SLC35G1 remains unknown. Since the SLC35G1-mediated uptake appears to be similar in the presence and absence of potassium, the authors argue that membrane potential has no role in the transport process. Since it is proposed that the divalent citrate is the substrate for the transporter, it is difficult to reconcile with the conclusion that the membrane potential has no impact on the transport process, especially given the fact that no other exchangeable anion has been shown or suggested. Even if chloride is the potential exchangeable anion, it still begs the question as to the stoichiometry of citrate:chloride if membrane potential plays no role. Obviously, additional work is needed to figure out the actual transport mechanism for SLC35G1.

<https://doi.org/10.7554/eLife.98853.2.sa2>

Reviewer #3 (Public review):

The authors convincingly show that SLC35G1 mediates uptake of citrate which is dependent on pH and chloride concentration. Putting their initial findings in a physiological context, they present human tissue expression data of SLC35G. Their Transwell assay indicates that SLC35G1 is a citrate exporter at the basolateral membrane.

Weaknesses:

The manuscript would benefit from the inclusion of the antibody validation results. Related to the localization of SLC35G1, the polyclonal antibody was not validated in the knockdown

cells used in the study. This would strengthen the antibody validation, the localization results as well as the transport assay in 2C.

Also, it is unclear why the Transwell assay was not performed upon knockdown of SLC35G1 to support the conclusions.

<https://doi.org/10.7554/eLife.98853.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

The current manuscript provides strong evidence that the molecular function of SLC35G1, an orphan human SLC transporter, is citrate export at the basolateral membrane of intestinal epithelial cells. Multiple lines of evidence, including radioactive transport experiments, immunohistochemical staining, gene expression analysis, and siRNA knockdown are combined to deduce a model of the physiological role of this transporter.

Strengths:

The experimental approaches are comprehensive, and together establish a strong model for the role of SLC35G1 in citrate uptake. The observation that chloride inhibits uptake suggests an interesting mechanism that exploits the difference in chloride concentration across the basolateral membrane.

Weaknesses:

Some aspects of the results would benefit from a more thorough discussion of the conclusions and/or model.

For example, the authors find that SLC35G1 prefers the dianionic (singly protonated) form of citrate, and rationalize this finding by comparison with the substrate selectivity of the citrate importer NaDC1. However, this comparison has weaknesses when considering the physiological pH for SLC35G1 and NaDC1. NaDC1 binds citrate at a pH of ~5.4 (the pKa of citrate is 5.4, so there is a lot of dianionic citrate present under physiological circumstances). SLC35G1 binds citrate under pH conditions of ~7.5, where a very small amount of dianionic citrate is present. The data clearly show a pH dependence of transport, and the authors rule out proton coupling, but the discrepancy between the pH dependence and the physiological expectations should be addressed/commented on.

Thank you for your insightful comment. Citrate exists mostly in its trianionic form under near neutral pH conditions in biological fluids, as you pointed out. Its dianionic form represents only a small portion (about 1/100) of total citrate due to the pKa. However, significant SLC35G1-specific uptake was observed under near neutral pH conditions (Figure 1G). Therefore, although SLC35G1-mediated citrate transport is less efficient under physiologically relevant near neutral pH conditions, it could still play a role particularly in the intestinal absorption process, in which the concentration gradient of dianionic citrate could be maintained by continuous supply by NaDC1-mediated apical uptake.

The rationale for the series of compounds tested in Figure 1F, which includes metabolites with carboxylate groups, a selection of drugs including anion channel inhibitors and statins, and bile acids, is not described. Moreover, the lessons drawn from this experiment are vague and should be expanded upon. It is not clear what, if anything, the compounds that reduce citrate uptake have in common.

Thank you for highlighting the need for clarity regarding the compounds tested in Figure 1F. The tested compounds were TCA cycle intermediates (fumarate, α -ketoglutarate, malate, pyruvate, and succinate) as substrate candidate carboxylates analogous to citrate, diverse anionic compounds (BSP, DIDS, probenecid, pravastatin, and taurocholate) as those that might be substrates or inhibitors, and diverse cationic compounds (cimetidine, quinidine, and verapamil) as those that are least likely to interact with SLC35G1. Among them, certain anionic compounds significantly reduced SLC35G1-specific citrate uptake, suggesting that they may interact with SLC35G1. However, we could not identify any structural features commonly shared by these compounds, except that they have anionic moieties. We acknowledge that it requires further elaboration to clarify such structural features. We have revised the relevant section on p. 3 (line 25 - 32) to include these.

The transporter is described as a facilitative transporter, but this is not established definitively. For example, another possibility could involve coupling citrate transport to another substrate, possibly even chloride ion.

Thank you for your insightful comment regarding the nature of SLC35G1's transport mechanism. While we have described SLC35G1 as a facilitative transporter based on our current data, we acknowledge that this has not been definitively proven, as you pointed out, and we cannot exclude the possibility that its sensitivity to extracellular Cl^- might imply its operation as a citrate/ Cl^- exchanger. To examine the possibility, we would need to manipulate the chloride ion gradient across the plasma membrane. Particularly, generating an outward Cl^- gradient to see if it could enhance citrate uptake could be a potential strategy. However, current techniques do not allow us to effectively generate the Cl^- gradient, thus preventing us from conclusively verifying this possibility. We recognize the importance of further investigating this aspect in future studies. Your suggestion highlights an important area for additional research to fully understand the transport mechanism of SLC35G1. We have additionally commented on this issue on p. 4 (line 1 - 3).

Reviewer #2 (Public Review):

Summary:

The primary goal of this study was to identify the transport pathway that is responsible for the release of dietary citrate from enterocytes into blood across the basolateral membrane.

Strengths:

The transport pathway responsible for the entry of dietary citrate into enterocytes was already known, but the transporter responsible for the second step remained unidentified. The studies presented in this manuscript identify SLC35G1 as the most likely transporter that mediates the release of absorbed citrate from intestinal cells into the serosal side. This fills an important gap in our current knowledge of the transcellular absorption of dietary citrate. The exclusive localization of the transporter in the basolateral membrane of human intestinal cells and the human intestinal cell line Caco-2 and the inhibition of the transporter function by chloride support this conclusion.

Weaknesses:

(i) The substrate specificity experiments have been done with relatively low concentrations of potential competing substrates, considering the relatively low affinity of the transporter for citrate. Given that NaDC1 brings in not only citrate as a divalent anion but also other divalent anions such as succinate, it is possible that SLC35G1 is responsible for the release of not only citrate but also other dicarboxylates. But the substrate specificity studies show that the dicarboxylates tested did not compete with citrate, meaning that SLC35G1 is selective for the citrate (2-), but this conclusion might be flawed because of the low concentration of the competing substrates used in the experiment.

Thank you for your valuable comment on our substrate specificity experiments. As you pointed out, we cannot rule out the possibility that dicarboxylates might be recognized by SLC35G1 with low affinity as the tested concentration was relatively low. However, at the concentration of 200 μ M, competing substrates with an affinity comparable to that of citrate could inhibit SLC35G1-specific citrate uptake by about 30%. Therefore, it is likely that the compounds that did not exhibit significant effect have no affinity or at least lower affinity than citrate to SLC35G1. Further studies should explore a broader range of concentrations for potential substrates including those with lower affinity. It would help clarify the substrate recognition characteristics of SLC35G1 and if it indeed has a unique preference for citrate over dicarboxylates. We have additionally mentioned that on p. 3, line 32 – 35.

(ii) The authors have used MDCK cells for assessment of the transcellular transfer of citrate via SLC35G1, but it is not clear whether this cell line expresses NaDC1 in the apical membrane as the enterocytes do. Even though the authors expressed SLC35G1 ectopically in MDCK cells and showed that the transporter localizes to the basolateral membrane, the question as to how citrate actually enters the apical membrane for SLC35G1 in the other membrane to work remains unanswered.

Thank you for highlighting this important aspect of our study. The mechanism of apical citrate entry in MDCKII cells is unknown, although NaDC1 or a similar transporter may be involved. However, this set of experiments have successfully demonstrated the basolateral localization of SLC35G1 and its operation for citrate efflux. Attempts to clarify the apical entry mechanism may need to be included in future studies for more detailed characterization of the model system using MDCKII cells. This would help in fully understanding the transcellular transport system for citrate. Investigation using Caco-2 cells or MDCKII cells double transfected with NaDC1 and SLC35G1 would also need to be induced in future studies to gain more definitive insights into the transcellular transport mechanism for citrate in the intestine, delineating the suggested cooperative role of NaDC1 and SLC35G1. We would be grateful for your understanding of our handling regarding this issue.

(iii) There is one other transporter that has already been identified for the efflux of citrate in some cell types in the literature (SLC62A1, PLoS Genetics; 10.1371/journal.pgen.1008884), but no mention of this transporter has been made in the current manuscript.

Thank you for bringing up the relevance of SLC62A1, which has recently been identified as a citrate efflux transporter in some cell types (PLoS Genet, 16, e1008884, 2020). We have now included comments on this transporter in Introduction (p. 2).

Reviewer #3 (Public Review):

Summary:

Mimura et al describe the discovery of the orphan transporter SLC35G1 as a citrate transporter in the small intestine. Using a combination of cellular transport assays, they show that SLC35G1 can mediate citrate transport in small intestinal cell lines. Furthermore, they investigate its expression and localization in both human tissue and cell lines. Limited evidence exists to date on both SLC35G1 and citrate uptake in the small intestine, therefore this study is an important contribution to both fields. However, the main claims by the authors are only partially supported by experimental evidence.

Strengths:

The authors convincingly show that SLC35G1 mediates uptake of citrate which is dependent on pH and chloride concentration. Putting their initial findings in a physiological context, they present human tissue expression data of SLC35G. Their Transwell assay indicates that SLC35G1 is a citrate exporter at the basolateral membrane.

Weaknesses:

Further confirmation and clarification are required to claim that the SLC indeed exports citrate at the basolateral membrane as concluded by the authors. Most experiments measure citrate uptake, but the authors state that SLC35G1 is an exporter, mostly based on the lack of uptake at physiological conditions faced at the basolateral side. The Transwell assay in Figure 1L is the only evidence that it indeed is an exporter. However, in this experiment, the applied chloride concentration was not according to the proposed model (120 mM at the basolateral side). The Transwell assay, or a similar assay measuring export instead of import, should be carried out in knockdown cells to prove that the export indeed occurs through SLC35G1 and not through an indirect effect. Related to the mentioned chloride sensitivity, it is unclear how the proposed model works if the SLC faces high chloride conditions under physiological conditions though it is inhibited by chloride.

Thank you for highlighting these important points. We used the Cl⁻-rich medium in transcellular transport studies, as stated in the relevant section in Materials and Methods (p. 6, line 2 – 5). The Cl⁻ concentration (144 mM) was comparable to the physiological concentration in extracellular body fluids. To clarify that experimental condition, we have additionally noted that in the text (p. 4, line 9) and the legends of Figs. 1K and 1L. The results indicate that basolaterally localized SLC35G1 can mediate citrate export effectively under the Cl⁻-rich extracellular condition. The transport mechanism regulated by Cl⁻ is unclear, but it is difficult to further clarify the mechanism at this time. We recognize the importance of further investigating the aspect in future studies, including the possibility that SLC35G1 might be a citrate/Cl⁻ exchanger, as pointed out by Reviewer #1 (3rd comment).

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

The figures are very tiny and difficult to see. The inset in Figure 1C is much too small to be readable. I suggest enlarging the panels.

Thank you for your feedback. As advised, we have enlarged the panels to improve visibility.

Line 74: "certain anionic compounds significantly inhibited SLC35G1-specific citrate uptake, indicating they are also recognized by SLC35G1." This sentence should be reworded since the mechanism is not clear. The word "reduced" would be a better option

than "inhibited." Are there other interpretations besides SLC35G1 binding to explain the observations?

Thank you for your suggestion. We have reworded the sentence to improve clarity (p. 3, line 30). It may be possible to speculate that they interact with SLC35G1, but the mechanisms are not clear yet.

The manuscript is vague about how the transporter was discovered. If a screen of orphan transporters was performed to identify a citrate transporter, this should be described.

Thank you for pointing out the need for more details regarding the discovery of the transporter. We have added some detailed description at the beginning of Results and Discussion (p. 3).

Reviewer #2 (Recommendations For The Authors):

Recommendations for the authors:

(1) For transcellular transport of citrate and the role of SLC35G1, it would be better to use Caco-2 cells cultured on Transwells because these cells express NaDC1 in the apical membrane and the authors have shown that SLC35G1 is expressed in the basolateral membrane in this cell line. The mechanism for the entry of citrate into MDCK cells used in the present manuscript is not known. If the authors prefer to use MDCK cells because of their superior use for polarization, they can use a double transfection (NaDC1 and SLC35G1) to differentially express the two transporters in the apical versus and basolateral membrane and then use the cells for trans cellular transport of citrate.

Please refer to our reply to your second review comment.

(2) The substrate specificity experiments should use concentrations higher than 0.2 mM for competing dicarboxylates because the K_m for citrate is only 0.5 mM. It is likely that NaDC1 brings in citrate and other dicarboxylates into enterocytes and then SLC35G1 mediates the efflux of these metabolic intermediates into blood.

Please refer to our reply to your first review comment.

(3) One major aspect of the transport function of this newly discovered citrate efflux transporter that has not been explored is the role of membrane potential in the transport function. The transporter is not coupled to Na or K or even H; so then the transport of citrate via this transporter must be electrogenic. Of course, this would be perfect for the transporter to function in the efflux of citrate because of the inside-negative membrane potential, but the authors need to show that the transporter is electrogenic. This can be examined through Caco-2 cells and/or MDCK cells expressing SLC35G1 and examining the impact of changes in membrane potential (valinomycin and K) on the transport of citrate.

Thank you for your suggestion. As shown in Figure 1D, the use of K-gluconate in place of Na-gluconate, which induces plasma membrane depolarization, had no impact on the specific uptake of citrate, suggesting that SLC35G1-mediated citrate transport is independent of membrane potential. We have additionally mentioned this on p. 3 (line 21 – 24).

(4) The localization studies mention Na/K ATPase component as a basolateral membrane marker, but the text describes it as BCRP. This needs to be corrected.

Thank you for pointing out the mistake. We have corrected that. The marker was ATP1A1.

Reviewer #3 (Recommendations For The Authors):

Major points:

- (1) Most experiments measure citrate uptake, but the authors state that SLC35G1 is an exporter, mostly based on the lack of uptake at physiological conditions faced at the basolateral side. The Transwell assay in Figure 1L is the only evidence that it indeed is an exporter. However, in this experiment, the applied chloride concentration was not according to the proposed model (120mM at basolateral side). Why was this chloride concentration not mimicked accordingly in the Transwell assay?*
- (2) The Transwell assay, or a similar assay measuring export instead of import, should be carried out in knockdown cells to prove that the export indeed occurs through SLC35G1 and not through an indirect effect.*
- (3) Related to the mentioned chloride sensitivity, it is unclear how the proposed model works if the SLC faces high chloride conditions under physiological conditions though it is inhibited by chloride.*

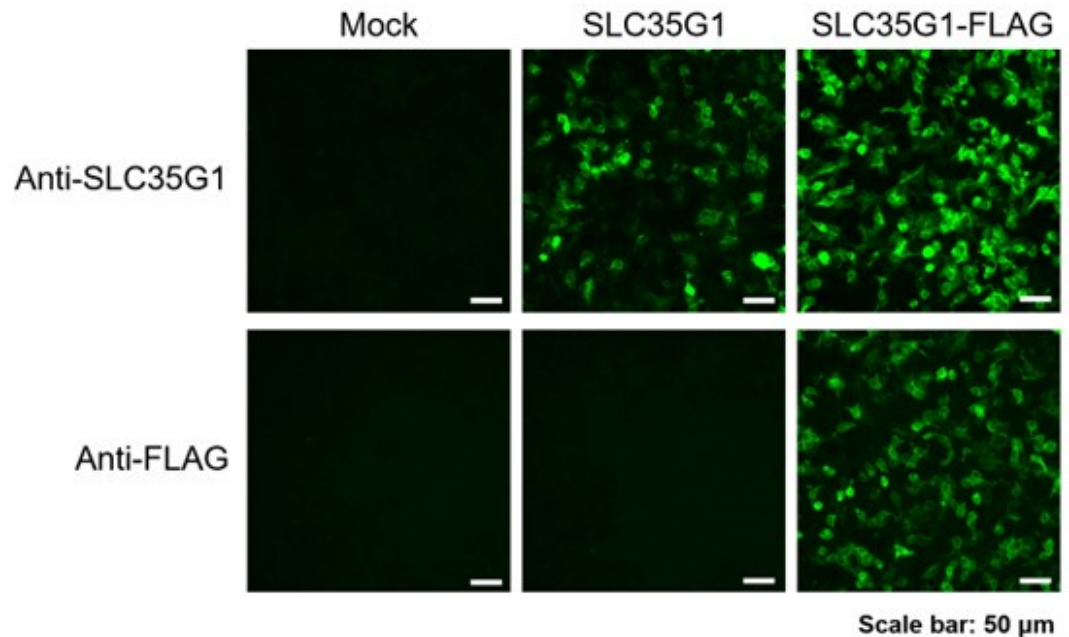
Please refer to our reply to your review comments.

Related to the localization of SLC35G1:

- (4) The polyclonal antibody against SLC35G1 should be validated to prove the specificity. This should be relatively straightforward given the authors have SLC35G1 knockdown cells.*

Thank you for your suggestion. To validate the specificity of the polyclonal antibody against SLC35G1, we prepared HEK293 cells transiently expressing SLC35G1 and SLC35G1 tagged with a FLAG epitope at the C-terminus (SLC35G1-FLAG). In the immunostained images, whereas only SLC35G1-FLAG was stained with the anti-FLAG antibody, both SLC35G1 and SLC35G1-FLAG were stained with the anti-SLC35G1 antibody, indicating that the anti-SLC35G1 antibody can recognize SLC35G1. In addition, the localization patterns of SLC35G1-FLAG observed with both antibodies were consistent, indicating furthermore that the anti-SLC35G1 antibody can recognize SLC35G1 specifically. Based on all these, the specificity of the anti-SLC35G1 antibody was validated.

Author response image 1.



(5) To strengthen the data on the localization of SLC35G1, the cell lines should be co-stained with a plasma membrane marker as well, not just in tissue with ATP1A1. In polarized cells co-staining with apical and basolateral markers should be applied.

SLC35G1 was indicated to be localized to the basolateral membrane geometrically in both polarized MDCKII and Caco-2 cells. This finding aligns with its basolateral localization indicated by its colocalization with ATP1A1 in the human small intestinal section. These results are we consider sufficient to support the basolateral localization characteristics of SLC35G1.

General points:

(6) In the abstract the authors mention that they focus on highly expressed orphan transporters in the small intestine as candidates. However, no other candidates are mentioned or discussed in the study. Consequently, this should be rephrased.

Thank you for the advice. Also taking into consideration the third recommendation point by Reviewer #1, we have added some detailed description at the beginning of Results and Discussion (p. 3).

(7) As far as mentioned there is exactly one (other) publication on SLC35G1 (10.1073/pnas.1117231108). The authors should discuss this only publication with functional data on SLC35G1 in more detail. How do the authors integrate their findings with the existing knowledge? For example, why did the authors not investigate the impact of Ca^{2+} on SLC35G1 transport?

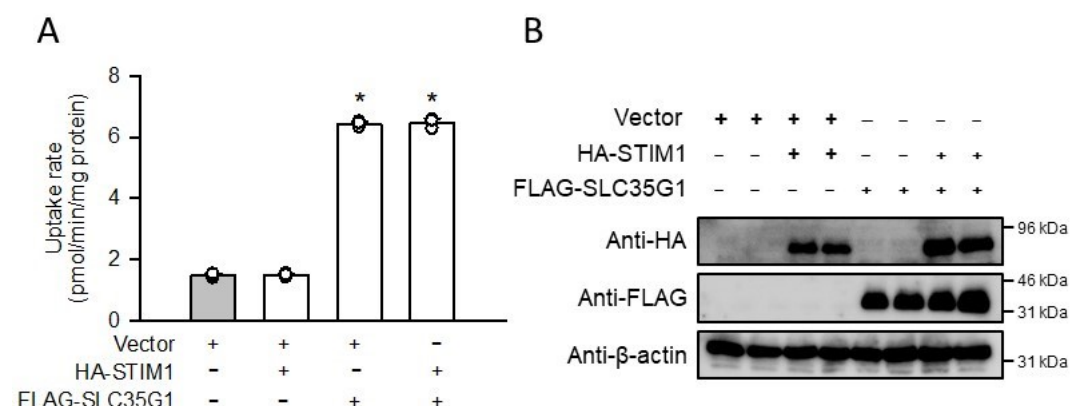
Thank you for your suggestion. SLC35G1 was indicated to be mainly localized to the endoplasmic reticulum (ER) in the earlier study, in which SLC35G1 was tagged with GFP. A possibility is that SLC35G1 was wrongly directed to ER due to the modulation in the study. We

have additionally mentioned this possibility in the relevant section (p. 3, line 9 – 11). We have also revised a relevant sentence on p. 3 (line 5).

With regard to another point that GFP-tagged SLC35G1 was indicated to interact with STIM1, we examined its effect on SLC35G1-mediated citrate uptake supplementary. As shown in the accompanying figure, coexpression of HA-tagged STIM1 did not affect the elevated citrate uptake induced by FLAG-tagged SLC35G1, indicating that STIM1 has no impact on citrate transport function of SLC35G1 at the plasma membrane.

Author response image 2.

(A) Effect of the coexpression of HA-tagged STIM1 on [14C]citrate (1 μ M) uptake by FLAG-tagged SLC35G1 transiently expressed in HEK293 cells. The uptake was evaluated for 10 min at pH 5.5 and 37°C. Data represent the mean $\pm$ SD of three biological replicates. Statistical differences were assessed using ANOVA followed by Dunnett's test. *, $p < 0.05$ compared with the control (gray bar). (B) Western blot analysis was conducted by probing for the HA and FLAG tags, using the whole-cell lysate samples (10 μ g protein aliquots) prepared from cells expressing HA-STIM1 and/or FLAG-SLC35G1. The blots of β -actin are shown for reference.



(8) Generally, the introduction could provide more background.

In response to your suggestion and also to the third review comment from Reviewer #2, we have now additionally included comments on SLC62A1, which has recently been reported as a citrate efflux transporter in some cell types, in Introduction.

Minor points:

(9) There is a typo in Figure 1D: mannitol instead of mannitol.

Thank you for pointing that out. We have corrected the typo in Figure 1D.

(10) Figure 1J: The resolution is low and the localization to the basolateral membrane is not conclusive based on this image. It seems rather localized at the whole membrane and intracellularly too.

Thank you for your feedback. We have enhanced the resolution of the image and also enlarged it to improve clarity and make the basolateral membrane localization more discernible.

(11) Figure 1K: Clarification is needed if the experiment was performed in the Transwell plate. Based on the results from the pH titration experiment, it is expected that there is no uptake at pH7.4. Therefore, this experiment does not seem to provide additional evidence or support the conclusions drawn related to cellular polarization.

Please refer to our reply to your review comments.

<https://doi.org/10.7554/eLife.98853.2.sa0>